Waterbirds Detection: Classifying a Minority

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Code: github.com/AleMdP/waterbirds-shortcut-learning

1. INTRODUCTION

Standard vision models trained via Empirical Risk Minimization (ERM) optimize the average loss over the training distribution. When non-causal background correlates strongly with object classes, gradient descent prioritizes opportunistic background features over robust semantic representations—a failure mode known as shortcut learning (Geirhos et al., 2020). In datasets with severe subgroup imbalance, ERM minimizes overall loss while collapsing on under-represented sub populations.

The Waterbirds benchmark (Sagawa et al., 2020) isolates this problem by pasting CUB bird photographs (Wah et al., 2011) onto Places backgrounds (Zhou et al., 2018) such that 95% of waterbirds appear on water and 95% of landbirds on land in the training set. This yields four groups (y, a) defined by bird class y and background a:

- Group 0 (y=0, a=0): **Landbird on Land** (Majority / Spurious, N_{train} = 3,498)
- Group 1 (y=0, a=1): **Landbird on Water** (Minority / Out-of-Distribution, N_{train} = 184)
- Group 2 (y=1, a=0): **Waterbird on Land** (Minority / Out-of-Distribution, N_{train} = 56)
- Group 3 (y=1, a=1): **Waterbird on Water** (Majority / Spurious, N_{train} = 1,057)

Because overall average accuracy masks subpopulation failure, the primary evaluation metric across all experiments is Worst-Group Accuracy ($Acc_{worst} = \min_g Acc_g$), supplemented by Average Accuracy (Acc_{avg}) to assess global performance trade-offs.

Sagawa et al. (2020) showed that Group DRO—which minimizes worst-case group loss—only helps over-parameterized networks when combined with regularization (strong weight decay or early stopping); otherwise, a network achieving zero training loss is already optimal for the worst-case objective. Our study addresses three questions: (i) how much worst-group gain stems from simply rebalancing the data (via sampler or loss weights) versus Group DRO itself; (ii) whether a Vision Transformer backbone behaves differently from a ResNet; and (iii) how much average accuracy robust methods sacrifice. We contribute a controlled 3 objectives × 2 samplers × 2 backbones

comparison, a 14-run Group DRO hyperparameter sweep with validation model selection, and an analysis of how loss reweighting interacts with balanced sampling.

2. METHODS AND MATERIALS

Dataset. We use the official Waterbirds split (Sagawa et al., 2020; Sagawa & Koh, 2020): 4,795 training, 1,199 validation, and 5,794 test images. Validation (467 / 466 / 133 / 133 images for G0–G3) and test sets (2,255 / 2,255 / 642 / 642) are group-balanced within each bird class to evaluate minority groups reliably.

Backbones. We fine-tune all layers of two torchvision models with ImageNet-1k pretrained weights and a new 2-class head: ResNet-50 (He et al., 2016) and ViT-B/16 (Dosovitskiy et al., 2021). The two checkpoints come from different training recipes, so backbone comparisons are indicative only.

Objectives and Data Loading. Models are trained with SGD (momentum 0.9, learning rate 10^{-4}, weight decay 10^{-4}, batch size 32, 5 epochs) using RandomResizedCrop(224, scale=(0.7, 1.0)) and random horizontal flips. We compare three training objectives:

- **ERM**: Standard mean cross-entropy loss over the mini-batch.
- **Group-reweighted loss**: Per-sample cross-entropy scaled by inverse group frequency $N / (4 \cdot n_g)$, weighting Group 2 by ≈21.4× and Group 0 by ≈0.34×.
- **Group DRO**: Following Algorithm 1 of Sagawa et al. (2020), we maintain a distribution q over groups, updated at each step via $q_g \leftarrow q_g \cdot \exp(\eta \cdot (\ell_g + C / \sqrt{n_g}))$ and renormalized, where ℓ_g is the mini-batch mean loss of group g, η is the step size, and C is the group-adjustment coefficient. The model optimizes $\sum_g q_g \ell_g$.

Each objective is paired with either (a) standard random shuffling (unbalanced) or (b) a group balanced WeightedRandomSampler with sample weight $1/n_g$ drawn with replacement (balanced). Pairing ERM with the balanced sampler implements the plain upweighting baseline of Sagawa et al. (2020), whereas pairing the reweighted loss with the balanced sampler applies the inverse-frequency correction twice.

Hyperparameter Sweep and Evaluation Protocol. On both backbones with the balanced sampler, we evaluate 14 Group DRO configurations: $\eta \in \{0.001, 0.01, 0.05, 0.1, 0.5, 1.0\}$; weight decay $\in \{10^{-3}, 10^{-2}, 10^{-1}\}$; adjustment $C \in \{1, 2\}$ with weight decay $\in \{10^{-2}, 10^{-1}\}$; and a higher learning rate run (C = 1, weight decay 10^{-3}, lr = 10^{-3}). For every run (single seed), we select the epoch with the highest validation worst-group accuracy (breaking ties by validation average accuracy) and evaluate once on the test set. We report worst-group accuracy (W), group-balanced test average accuracy (Avg), and average accuracy weighted by training group proportions (Avg-w), which reflects accuracy under the natural skewed distribution (Sagawa et al., 2020). Parts of the codebase were written with generative AI assistance (Gemini) and verified by the authors.

3. RESULTS

3.1 Main Comparison Across Objectives, Samplers, and Backbones

Table 1: Test accuracy (%) after 5 epochs (one run per row). W: worst-group accuracy; Avg: group-balanced test average; Avg-w: average weighted by training group frequencies; G0–G3: per-group accuracies.

Backbone	Method	Sampler	W	Avg	Avg-w	G0	G1	G2	G3
ResNet-50	ERM	unbalanced	8.88	73.33	93.27	99.69	62.93	8.88	81.78
ResNet-50	Reweight	unbalanced	82.93	87.19	90.44	91.09	82.93	85.83	89.88
ResNet-50	Group DRO	unbalanced	61.53	84.19	95.48	98.45	74.41	61.53	91.12
ResNet-50	ERM	balanced	84.08	86.62	88.39	88.43	84.08	86.76	89.10
ResNet-50	Reweight	balanced	74.90	85.14	77.33	74.90	93.84	94.08	81.62
ResNet-50	Group DRO	balanced	**86.74**	88.92	90.74	91.18	86.74	87.38	90.19
ViT-B/16	ERM	unbalanced	51.09	83.34	96.55	99.56	73.57	51.09	92.99
ViT-B/16	Reweight	unbalanced	**88.63**	90.56	91.95	92.73	89.09	88.63	90.03
ViT-B/16	Group DRO	unbalanced	82.35	89.44	95.67	97.43	82.35	82.87	92.83
ViT-B/16	ERM	balanced	81.31	92.98	95.86	98.00	92.06	81.31	90.19
ViT-B/16	Reweight	balanced	88.32	91.65	91.68	92.33	92.51	88.32	89.56
ViT-B/16	Group DRO	balanced	84.42	90.99	94.99	96.36	87.10	84.42	92.37

Standard ERM exhibits severe shortcut learning (Table 1): ResNet-50 reaches 99.69% accuracy on Group 0 but collapses to 8.88% on Group 2 (waterbirds on land). ViT-B/16 is less affected under ERM (51.09% on Group 2) but still lags behind majority groups by over 40 points. Every mitigation raises worst-group accuracy substantially, reaching 61.53–86.74% on ResNet-50 and 81.31–88.63% on ViT-B/16. Plain upweighting via the balanced sampler with ERM accounts for the vast majority of the improvement, reaching 84.08% (ResNet-50) and 81.31% (ViT-B/16). Adding Group DRO on top of the balanced sampler yields only +2.66 and +3.11 points, respectively, while running Group DRO without the balanced sampler drops ResNet-50 worst-group accuracy to 61.53%.

Peak test worst-group accuracy is 86.74% for ResNet-50 (Group DRO, balanced) and 88.63% for ViT-B/16 (Reweight, unbalanced). However, validation rankings differ slightly (led by ERM + balanced at 84.5% on ResNet-50, and Reweight + balanced at 87.2% on ViT-B/16), indicating that differences among top robust methods fall within single-run evaluation noise (≈1.4 percentage points binomial standard error on 642 test images, and ≈3.0 points on 133 validation images). Conversely, combining the balanced sampler with the reweighted loss clearly over-corrects on ResNet-50 (not on Vit-B/16): minority accuracies rise to 93.84% (G1) and 94.08% (G2), while majority Group 0 drops to 74.90%, pulling worst-group accuracy down to 74.90%.

3.2 Robustness vs. Average Accuracy and Training Loss Dynamics

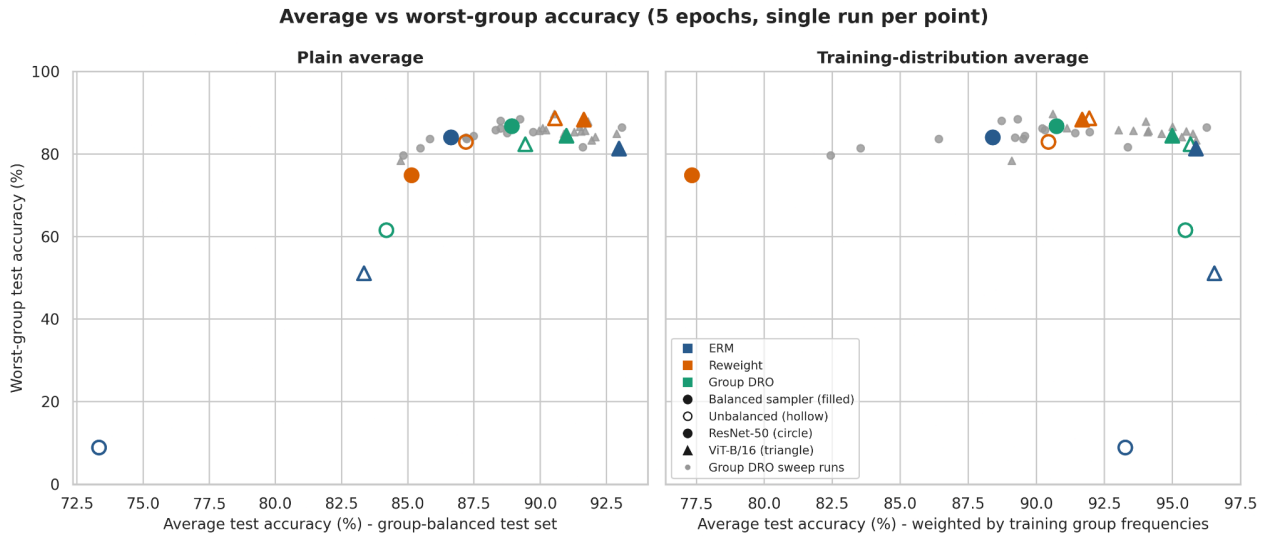


Figure 1: Worst-group vs. average test accuracy across all 5-epoch runs. Left: unweighted group-balanced test average (Avg). Right: average weighted by training group frequencies (Avg-w). Grey markers denote Group DRO sweep runs.

Whether robustness costs average accuracy depends on the evaluation distribution (Figure 1). On the group-balanced test set (Avg), robust methods strictly improve upon unbalanced ERM (73.33% on ResNet-50) because minority-group failures heavily penalize the unweighted mean. Under the natural training distribution (Avg-w), unbalanced ERM achieves the highest accuracy (93.27% on ResNet-50; 96.55% on ViT-B/16), and switching to the best worst-group methods incurs a modest 2.53 to 4.60 point reduction in Avg-w in exchange for +37.5 to +77.9 points in worst-group accuracy.

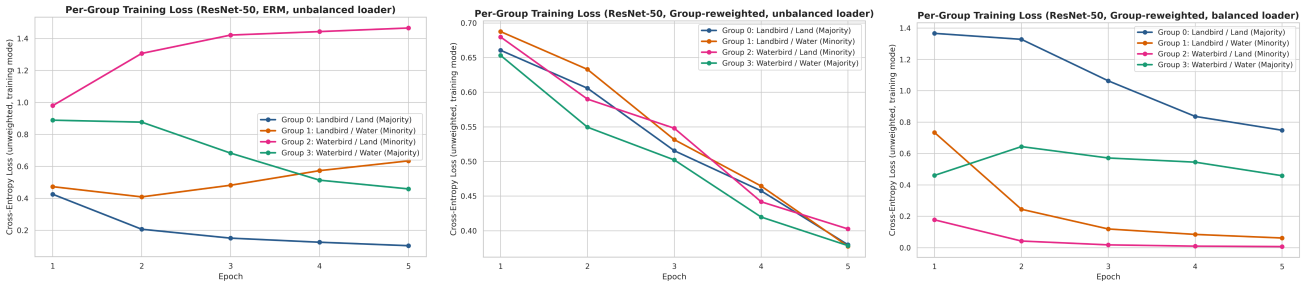


Figure 2: Per-group unweighted cross-entropy loss during training for ResNet-50 under (left) ERM unbalanced, (center) Reweight unbalanced, and (right) Reweight balanced (double-weighted).

The ResNet-50 training loss trajectories (Figure 2) directly illustrate the mechanics of shortcut learning and over-correction. Under unbalanced ERM, majority losses decrease steadily (G0: 0.43 → 0.10; G3: 0.89 → 0.46) while minority losses actively diverge (G1: 0.47 → 0.63; G2: 0.98 → 1.47, with

8.9% epoch-5 training accuracy). Applying inverse-frequency loss reweighting on the unbalanced loader aligns all four groups onto a shared downward trajectory (0.65–0.69 → 0.38–0.40). However, combining reweighting with the balanced sampler (Figure 2, right) drives Group 2 loss down to 0.01 while leaving majority Group 0 elevated at 0.75, explaining its 74.90% test collapse.

3.3 Group DRO Hyperparameter Sweep

Table 2: Group DRO hyperparameter sweep (balanced sampler, 5 epochs). VW: validation worst-group accuracy of the selected epoch; W and Avg: test worst-group and group-balanced average accuracy (%). Bold indicates validation-selected configurations.

Configuration	ResNet VW	ResNet W	ResNet Avg	ViT VW	ViT W	ViT Avg
$\eta = 0.001$	86.5	85.76	88.32	85.0	84.89	92.89
$\eta = 0.01$ (default)	83.5	85.06	88.75	84.2	85.36	91.28
$\eta = 0.05$	85.7	83.68	87.23	85.0	85.83	90.21
$\eta = 0.1$	84.3	83.99	87.19	84.8	85.67	89.96
$\eta = 0.5$	85.0	83.64	85.83	82.7	83.33	91.94
$\eta = 1.0$	78.2	81.62	91.61	87.2	89.72	90.52
wd = 10^{-3}	84.2	86.16	88.51	85.7	85.51	91.54
wd = 10^{-2}	82.0	85.36	89.73	82.7	84.89	90.90
wd = 10^{-1}	84.2	84.35	87.49	84.2	84.11	92.08
C = 1, wd = 10^{-2}	85.7	88.47	89.23	84.2	86.60	91.49
C = 1, wd = 10^{-1}	85.7	88.03	88.51	85.0	85.67	91.70
C = 2, wd = 10^{-2}	80.5	79.69	84.83	**87.2**	**87.85**	**91.80**
C = 2, wd = 10^{-1}	82.2	81.42	85.47	85.7	86.29	90.09
C = 1, wd = 10^{-3}, lr = 10^{-3}	**87.2**	**86.45**	**93.08**	79.0	78.35	84.73

Across the 14 sweep configurations (Table 2), test worst-group accuracy spans 79.69–88.47% on ResNet-50 and 78.35–89.72% on ViT-B/16. Validation selection picks C = 1, wd = 10^{-3}lr = 10^{-3} for ResNet-50 (86.45% W, 93.08% Avg, 96.3% Avg-w) and C = 2, wd = 10^{-2} for ViT-B/16 (87.85% W, 91.80% Avg, winning the tie with η = 1.0 on validation average). Neither improves upon the best Table 1 configuration beyond evaluation noise. Notably, raising the learning rate to 10^{-3} yields the best validation model on ResNet-50 (simultaneously improving both worst-group and average accuracy over 5-epoch ERM) but degrades ViT-B/16 to 78.35%, highlighting strong backbone– learning rate coupling.

4. DISCUSSION

Rebalancing accounts for most of the robustness gain. Our experiments confirm that ERM relies heavily on background shortcuts when trained on the skewed Waterbirds distribution, failing even to fit minority Group 2 during training. However, simply equalizing group exposure (via a balanced sampler or loss reweighting) closes nearly the entire performance gap.

Consistent with Sagawa et al. (2020), Group DRO outperforms plain upweighting by only 2–3 percentage points, a margin comparable to single-run noise. Because our 5-epoch schedule keeps training cross-entropy around 0.36–0.40 rather than zero, our experiments operate squarely within the early-stopping regime where Sagawa et al. (2020) and Byrd & Lipton (2019) showed importance weighting and DRO remain effective without extreme weight decay. Indeed, our ResNet-50 progression from 8.88% (ERM) to 86.74% (Group DRO balanced) closely mirrors Sagawa et al.'s early-stopping benchmark (6.7% to 86.0%).

On the other hand, VIT-B/16, a modern representative of transformer models, appears to get worse results with Group DRO (even though not by much, around 4 points). This brings a question whether online Group DRO doesn’t actually interfere with the internal attention structure of the model, or what is behind this behaviour. But answering this question is outside of the scope of this work.

Why Group DRO requires balanced sampling, while reweighting must be applied once. Under standard shuffling, Group 2 (n = 56) is absent from roughly 69% of size-32 mini-batches. Because an absent group contributes zero loss to the multiplicative update of q, its adversarial weight is systematically underestimated whenever majority groups are present, explaining why unbalanced Group DRO lags on ResNet-50 (61.53% vs. 86.74%). Combining static inverse-frequency loss weights with a balanced sampler squares the frequency correction, starving majority Group 0 of gradient updates on ResNet-50. Thus, uniform group sampling is essential for online Group DRO updates for ResNet, but is not that required for ViT-B/16 which performs best just with static reweighting and adding balanced sampler doesn’t cause the model to overfit on the small groups like ResNet-50 does.

Backbone sensitivity, validation noise, and limitations. While ViT-B/16 exhibits substantially higher baseline robustness under ERM (51.09% vs. 8.88%) and greater resilience to double-weighting, this advantage reflects both architecture and pretraining differences. Furthermore, because each validation minority group contains only 133 images (0.75% per image), validation rankings fluctuate across hyperparameter settings (for instance, η = 1.0 scores 78.2% on ResNet-50

validation but 89.72% on ViT-B/16 test), leaving C = 2 on ViT-B/16 as the only consistent sweep signal matching Sagawa et al. (2020). Key limitations of this study include: (i) evaluating only the 5-epoch early-stopping regime rather than converged 300-epoch training with strong weight decay; (ii) running a single random seed per configuration; and (iii) limited parameter sweep that doesn't fully cover all combinations and doesn't explore some of the observed trends into detail.

Conclusion. On Waterbirds, mitigating shortcut learning primarily requires balancing group exposure via either a balanced data loader or inverse-frequency loss weighting but not both. For modern architectures (like ViT-B/16), Group DROs advantages are small to negligible.

4.1 References

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